Differentiation

ANSWER KEY

Section 1: Product & Quotient Rules

1 Differentiate $y = x^3e^{2x}$ with respect to x.

a. $x^2e^{2x}(3 + 2x)$ ✓

b. $3x^2e^{2x}$

c. $2x^3e^{2x}$

d. $x^2e^{2x}(3x + 2)$

Product rule: $dy/dx = (3x^2)e^{2x} + x^3(2e^{2x}) = x^2e^{2x}(3 + 2x)$.

2 Differentiate $y = \dfrac{\ln x}{x^2}$ for $x > 0$.

a. $\dfrac{1 - 2\ln x}{x^3}$ ✓

b. $\dfrac{1 + 2\ln x}{x^3}$

c. $\dfrac{1 - 2\ln x}{x^4}$

d. $\dfrac{1}{x^3}$

Quotient rule: $\frac{(1/x)(x^2) - (\ln x)(2x)}{x^4} = \frac{x - 2x\ln x}{x^4} = \frac{1 - 2\ln x}{x^3}$.

3 Find the gradient of the curve $y = x\sin(2x)$ at $x = \pi/4$.

a. **1** ✓

b. 0

c. $\pi/2$

d. 2

$dy/dx = \sin(2x) + 2x\cos(2x)$. At $x = \pi/4$: $\sin(\pi/2) + 2(\pi/4)\cos(\pi/2) = 1 + (\pi/2)(0) = 1$.

Section 2: Derivatives of Exponential, Logarithmic & Trig Functions

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4 Differentiate $f(x) = \ln(\sec x + \tan x)$.

a. $\sec x$ ✓

b. $\tan x$

c. $\sec^2 x$

d. $\sec x \tan x$

$f'(x) = \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} = \frac{\sec x(\tan x + \sec x)}{\sec x + \tan x} = \sec x$.

5 Differentiate $y = e^{\sin(3x)}$ with respect to x.

a. $3\cos(3x)e^{\sin(3x)}$ ✓

b. $\cos(3x)e^{\sin(3x)}$

c. $3e^{\sin(3x)}$

d. $-3\cos(3x)e^{\sin(3x)}$

Chain rule: $d/dx[e^{\sin(3x)}] = e^{\sin(3x)} \cdot d/dx[\sin(3x)] = 3\cos(3x)e^{\sin(3x)}$.

6 Find $\frac{d}{dx}[\tan^{-1}(2x)]$.

a. $\frac{2}{1+4x^2}$ ✓

b. $\frac{1}{1+4x^2}$

c. $\frac{2}{1+2x^2}$

d. $\frac{2}{\sqrt{1-4x^2}}$

$\frac{d}{dx}[\arctan(u)] = \frac{u'}{1+u^2} = \frac{2}{1+(2x)^2} = \frac{2}{1+4x^2}$.

Section 3: Implicit Differentiation

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7 Find $\frac{dy}{dx}$ for the curve $x^2 + xy + y^2 = 7$.

a. $-\frac{2x+y}{x+2y}$ ✓

b. $\frac{2x+y}{x+2y}$

c. $-\frac{2x}{2y}$

d. $-\frac{x+2y}{2x+y}$

$2x + (y + x dy/dx) + 2y dy/dx = 0 \Rightarrow dy/dx(x + 2y) = -(2x + y) \Rightarrow dy/dx = -\frac{2x+y}{x+2y}$.

8 Find the gradient of the curve $x^3 + y^3 = 9xy$ at the point $(2, 4)$.

a. $\frac{4}{5}$ ✓

b. $-\frac{4}{5}$

c. $\frac{5}{4}$

d. 1

$3x^2 + 3y^2y' = 9(y + xy') \Rightarrow x^2 + y^2y' = 3y + 3xy'$. At $(2, 4)$:
$4 + 16y' = 12 + 6y' \Rightarrow 10y' = 8 \Rightarrow y' = 8/10 = 4/5$.

9 Find the equation of the normal to the curve $x^2 - 2xy + 3y^2 = 9$ at $(3, 2)$.

a. $y = 2$ ✓

b. $x = 3$

c. $y = x - 1$

d. $2x + y = 8$

Differentiate: $2x - 2(y + xy') + 6yy' = 0$. At $(3, 2)$:
$6 - 2(2 + 3y') + 12y' = 0 \Rightarrow 6 - 4 - 6y' + 12y' = 0 \Rightarrow 2 + 6y' = 0 \Rightarrow y' = -1/3$. Normal slope is 3.
Equation: $y - 2 = 3(x - 3) \Rightarrow y = 3x - 7$.

Section 4: Parametric Differentiation

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10 A curve is defined parametrically by $x = 2t^2, y = 4t$. Find $\dfrac{dy}{dx}$ in terms of t.

a. $\dfrac{1}{t}$ ✓

b. t

c. $\dfrac{2}{t}$

d. $2t$

$dx/dt = 4t, dy/dt = 4$. Then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{4}{4t} = \frac{1}{t}$.

11 A curve has parametric equations $x = a\cos\theta, y = b\sin\theta$. Find the equation of the tangent at parameter θ.

a. $\dfrac{x}{a}\cos\theta + \dfrac{y}{b}\sin\theta = 1$ ✓

b. $\dfrac{x}{a}\sin\theta + \dfrac{y}{b}\cos\theta = 1$

c. $\dfrac{x}{a}\cos\theta - \dfrac{y}{b}\sin\theta = 1$

d. $bx\cos\theta + ay\sin\theta = ab$

$dx/d\theta = -a\sin\theta, dy/d\theta = b\cos\theta \Rightarrow dy/dx = -\frac{b\cos\theta}{a\sin\theta}$. Tangent:
$y - b\sin\theta = -\frac{b\cos\theta}{a\sin\theta}(x - a\cos\theta) \Rightarrow \frac{y\sin\theta}{b} - \sin^2\theta = -\frac{x\cos\theta}{a} + \cos^2\theta \Rightarrow \frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$.

12 Find $\dfrac{d^2y}{dx^2}$ for the parametric curve $x = t^2, y = t^3$.

a. $\dfrac{3}{4t}$ ✓

b. $\dfrac{3}{2}$

c. $\dfrac{3}{4t^2}$

d. $\dfrac{6}{t}$

$dx/dt = 2t, dy/dt = 3t^2 \Rightarrow dy/dx = 3t/2$. $\frac{d^2y}{dx^2} = \frac{d}{dt}[dy/dx]/\frac{dx}{dt} = \frac{3/2}{2t} = \frac{3}{4t}$.

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